

AXIOM D1
THE FUTURE OF SANITATION

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Abstract:

Urban drainage systems in Small Island Developing States, such as Trinidad and Tobago, regularly experience blockages due to plastic waste, organic debris and algae. This leads to residential flooding, property damage and public health concerns. Current upkeep involves manual labour, exposing workers to a hazardous environment. This report proposes AXIOM D1, a compact, amphibious robot designed to navigate shallow (0-15 cm depth), narrow (≤ 0.45 m), concrete drains. The robot collects solid waste via a servoactuated scoop and conveyor into a sealed 5 L container, scrubs algae with two 3000 RPM rotating nylon brushes, and streams live video to a mobile app via Wi-Fi for remote operation. The prototype targets the UN's sixth SDG – Clean Water and Sanitation, specifically Targets 6.3, 6.a and 6.b. The AXIOM D1 aims to reduce human exposure to contaminated water, enable automated means of cleaning and provide an affordable (\$358 USD) and scalable advancement for SIDS.

Introduction:

In the Caribbean, concrete "box" drains frequently clog with debris and algae, causing localized flooding and breeding disease-spreading mosquitoes. Currently, sanitation workers manually clear these narrow channels, a labor-intensive and infrequent task that exposes them to toxic hazards and physical injury. Because the drains are too shallow for heavy machinery yet too dangerous for humans, robotics is the ideal intervention. This paper proposes the AXIOM D1, an affordable, scalable, and trackable amphibious robot that routinely removes solid waste and algae from urban drains to reduce flooding, protect public health, and safeguard workers.

Problem Definition and Requirements:

MEASURABLE GOALS:

Metric	Target
Debris collection rate	≥ 0.5 kg per 100 m pass
Algae removal coverage	$\geq 80\%$ of contacted surface per pass
Operating speed (in drain)	0.10 – 0.20 m/s
Continuous battery life	≥ 60 minutes (realistic duty cycle)
Drain length per charge	≥ 400 m
Maximum operating water depth	15 cm
Deployment time (unpack to run)	≤ 5 minutes
System mass	≤ 15 kg
Chassis width (including tracks)	≤ 0.40 m
Wireless control range	≥ 50 m line-of-sight

CONSTRAINTS AND ASSUMPTIONS:

- **Cost:** Prototype budget is capped at USD \$500. The Bill of Materials totals \$358 (major costs: \$70 tracked chassis, \$55 LiPo battery, \$10 ESP32 CAM with WiFi and camera), leaving a \$142 contingency.

- **Environment:** Designed for tropical drains in Trinidad and Tobago (25–35°C, high humidity) with an operating depth of 0–15 cm. An aluminum 6061-T6 chassis resists corrosion, and electronics are sealed in an IP67 ABS enclosure.

- **Safety:** Teleoperated up to 50 m via WiFi and a live camera feed to keep workers out of hazardous drains. Safeguards include an emergency stop, a 15 A fuse, and ESP32 interlocks that halt motors if tilt exceeds 30°, water depth surpasses 15 cm, WiFi drops, or obstacles are detected.

- **Energy:** Fully battery-powered (11.1V, 10Ah LiPo) with no mains access. Average draw is 100.5 W, yielding a 56-minute runtime; upgrading to a 12Ah battery (+\$15) achieves the 60-minute target. The BMS provides thermal, overcharge, and deep-discharge protection.

- **Size and Weight:** Assembled dimensions are $400 \times 350 \times 150$ mm, weighing 14 kg (under the 15 kg portability limit). The ≤ 0.40 m width ensures it fits standard 0.45 m drains with 50 mm clearance.

- **Regulations:** Complies with local T&T drainage frameworks (MoWT, Municipal Corporations, WASA) as a non-invasive device. It adheres to international standards IEC 60529 (IP67), IEC 62133 (battery safety), AASHTO drain dimensions, and biofilm removal speed thresholds (≥ 5 m/s).

ENGINEERING REQUIREMENTS:

Requirement	Type
System shall navigate channels ≤ 0.40 m wide without wall contact under normal operation.	Functional
System shall collect ≥ 0.5 kg of solid debris per 100 m pass.	Functional
Brush assemblies shall achieve $\geq 80\%$ algae removal on contacted drain surfaces per pass.	Functional
Drive system shall maintain 0.10–0.20 m/s in up to 15 cm of standing water.	Functional
Battery system shall sustain full operation for ≥ 60 minutes under realistic duty cycle.	Functional
System shall trigger automatic stop if water depth exceeds 15 cm.	Functional
All electronics enclosures shall meet IP67 (IEC 60529).	Non-functional
E-stop shall halt all motors within 500 ms of activation.	Non-functional
Total assembled mass shall not exceed 15 kg.	Non-functional
WiFi control link shall maintain stable operation over ≥ 50 m line-of-sight.	Non-functional
Prototype unit cost shall not exceed USD \$500.	Non-functional
System shall be deployable (unpack to operational) within 5 minutes.	Non-functional

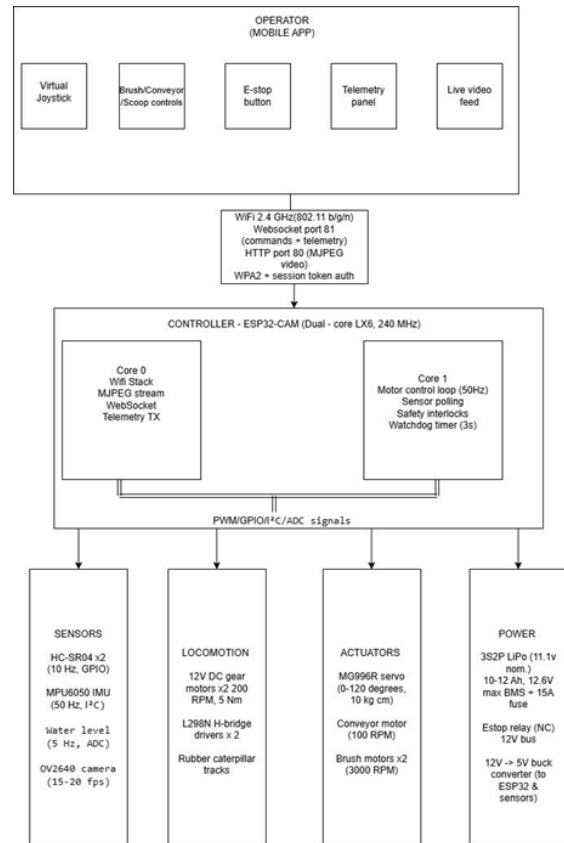
SYSTEM ARCHITECTURE AND

METHODOLOGY:

AXIOM D1 integrates six subsystems—locomotion, collection, cleaning, sensing, control/communication, and power within a command sense actuated pipeline. The ESP32-CAM processes operator commands over a 2.4 GHz WiFi WebSocket link, uses sensor

CONVERT GOALS INTO TECHNICAL SPECS

Signal flow: Operator → WiFi → ESP32-CAM
Controller → PWM/GPIO/I²C/ADC → Actuators
& Sensors → Physical drain maintenance. Sensor
data and telemetry flow in reverse: Sensors →
ESP32 → WiFi → Mobile App. E-stop signal path
is hardware only and bypasses firmware.



LOCOMOTION SUBSYSTEM

COLLECTION SUBSYSTEM

CLEANING SUBSYSTEM

SENSING AND PRECEPTION SUBSYSTEM

CONTROL AND COMMUNICATION SUBSYSTEM

Parameter	Value
Type	12 V DC gear motor with metal gearbox
No load speed	2600 RPM
Stall torque	190 Nm
Rated torque	7 Nm at 3 A
Gears	4 mm 6 profile
Speed calculation:	10-tooth sprocket, 42-tooth chain, 1:1 ratio
Trans sprocket pitch radius:	$r = 90 \text{ mm} \pm 0.01 \text{ mm}$
No load linear speed:	$v = 2\pi \cdot 2600 \cdot \frac{r}{60} = 25.13 \text{ m/s}$ or 56.2 km/h
Rated linear speed:	$v = 2\pi \cdot 1000 \cdot \frac{r}{60} = 10.5 \text{ m/s}$ or 23.3 km/h
This satisfies requirement R5 10, 10-10, 10-10, operating speed.	

5E. Safety & Reliability

Discuss: Hazards, Protections, Failsafes

Hazards: Environmental risks include submersion (>15 cm), tipping on debris, and confined collisions. Biological/chemical threats (leptospirosis, dengue, runoff) are eliminated via teleoperation. Electrical risks involve overcurrent, thermal runaway, and deep-discharge.

Protections (Hardware): A normally closed E-stop relay cuts the 12 V bus in <500 ms, bypassing firmware. A 15 A fuse handles overcurrent. The BMS ensures overcharge, 9.0 V deep-discharge, and 65 °C thermal cutoff. Electronics use an IP67 ABS enclosure, and cybersecurity relies on WPA2-PSK WiFi plus 128-bit session tokens.

Failsafes (Software Interlocks on Core 1):

Auto-stop triggers if ultrasonic distance <5 cm, IMU tilt >30°, or water level >15 cm. A 3-second watchdog halts motors on WebSocket signal loss, with 2-second auto-reconnects.

6. Modelling, Analysis & Simulation

6A. Engineering Calculations

Calculation 1: Traction Analysis

Objective: Verify that the tracks generate sufficient tractive force to propel the robot while pushing debris through standing water.

Given:

Robot mass: $m = 14 \text{ kg}$
Gravitational acceleration: $g = 9.81 \text{ m/s}^2$
Robot weight: $W = m \cdot g = 14 \times 9.81 = 137.3 \text{ N}$
Coefficient of friction (rubber on wet concrete): $\mu = 0.50$ [5]

Maximum tractive force:

$$F_{\text{traction}} = \mu \times W = 0.50 \times 137.3 = 68.7 \text{ N}$$

Resistance forces:

(a) Rolling resistance:

$$C_{\text{rr}} = 0.02 \text{ (rubber tracks on concrete)}$$
$$F_{\text{roll}} = C_{\text{rr}} \times W = 0.02 \times 137.3 = 2.75 \text{ N}$$

(b) Debris pushing (worst case: 2 kg object, $\mu_{\text{debris}} = 0.6$):

$$F_{\text{debris}} = \mu_{\text{d}} \times m_{\text{d}} \times g = 0.6 \times 2 \times 9.81 = 11.77 \text{ N}$$

(c) Hydrodynamic drag (in 10 cm standing water):

$$\text{Frontal area submerged: } A = 0.35 \times 0.10 = 0.035 \text{ m}^2$$
$$\text{Velocity: } v = 0.15 \text{ m/s}$$
$$\text{Drag coefficient: } C_{\text{d}} = 1.0 \text{ (bluff body)}$$

$$F_{\text{drag}} = 0.5 \times \rho \times v^2 \times C_{\text{d}} \times A$$
$$= 0.5 \times 1000 \times 0.0225 \times 1.0 \times 0.035$$
$$= 0.394 \text{ N}$$

Total resistance and safety factor:

$$F_{\text{total}} = 2.75 + 11.77 + 0.394 = 14.91 \text{ N}$$
$$\text{Safety factor} = F_{\text{traction}} / F_{\text{total}} = 68.7 / 14.91 = 4.61$$

Result: A safety factor of 4.6 confirms ample traction margin. Even on very slimy surfaces ($\mu = 0.3$):

$$F_{\text{traction}} = 0.3 \times 137.3 = 41.2 \text{ N}$$
$$\text{SF} = 41.2 / 14.91 = 2.76 \text{ — still adequate.}$$

Calculation 2: Motor Torque Verification

Objective: Verify that each track motor provides sufficient torque to overcome all resistance forces with margin.

Given:

Total resistance force (from Calc. 1): $F_{\text{total}} = 14.91 \text{ N}$

Number of drive motors: $n = 2$

Force per motor: $F_{\text{motor}} = 14.91 / 2 = 7.46 \text{ N}$

Track sprocket pitch radius: $r = 0.02 \text{ m}$

Required torque per motor:

$$T_{\text{required}} = F_{\text{motor}} \times r = 7.46 \times 0.02 = 0.149 \text{ Nm}$$

Available torque (rated):

$$T_{\text{rated}} = 5 \text{ Nm (at 3 A, per motor datasheet)}$$

Torque safety factor:

$$\text{SF} = T_{\text{rated}} / T_{\text{required}} = 5.0 / 0.149 = 33.6$$

Result: The motors are substantially over-specified for steady-state operation. The large torque margin accommodates transient loads: startup acceleration, climbing debris piles, and traversing drain joints or cracks. Even at 50% motor efficiency (accounting for gearbox and chain losses), the available torque of 2.5 Nm provides a safety factor of 16.8.

Calculation 3: Brush Effectiveness

Objective: Verify that brush tip speed exceeds the minimum threshold for effective biofilm removal from concrete surfaces.

Given:

Brush diameter: $d = 60 \text{ mm} = 0.06 \text{ m}$

Motor speed: $n = 3000 \text{ RPM}$

Brush tip speed:

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$$\begin{aligned} v_{\text{tip}} &= \pi \times d \times (n / 60) \\ &= \pi \times 0.06 \times (3000 / 60) \\ &= \pi \times 0.06 \times 50 \\ &= 9.42 \text{ m/s} \end{aligned}$$

'''

Minimum effective tip speed for biofilm removal: 5 m/s [5]

Safety factor:

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$$SF = 9.42 / 5.0 = 1.88$$

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Coverage rate estimate:

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Brush width (effective contact): $w = 30 \text{ mm} = 0.03 \text{ m}$

Robot forward speed: $v = 0.15 \text{ m/s}$

Number of brushes: 2

Cleaned strip width per pass: $2 \times 0.03 = 0.06 \text{ m}$

Drain floor width: 0.45 m

Passes for full coverage: $0.45 / 0.06 = 7.5 \rightarrow 8 \text{ passes (4 out + 4 return)}$

'''

Result: The brushes operate at 1.88x the minimum effective tip speed, providing adequate margin for worn bristles or reduced motor voltage under load. Full drain floor coverage requires approximately 4 round-trip passes within a single battery charge (56-67 min realistic).

Calculation 4: Collection Bin Capacity

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Bin volume: $5 \text{ L} = 0.005 \text{ m}^3$

Average debris density (mixed plastic + organic, loose): $\sim 400 \text{ kg/m}^3$

Bin capacity by mass: $0.005 \times 400 = 2.0 \text{ kg}$

Typical drain debris density: 5-15 g per linear metre.

At 10 g/m: $2000 / 10 = 200 \text{ m}$ per emptying.

At 5 g/m: $2000 / 5 = 400 \text{ m}$ per emptying.

'''

The bin requires 1-2 emptyings per charge cycle, each taking ~ 30 seconds at a drain access point.

6B. Simulation and Testing

Gazebo Simulation Environment

A physics-based simulation in Gazebo Classic 11.15.1 (Docker/Arch Linux) validated locomotion, sensors, and operational feasibility pre-prototype.

Robot Model: SDF 1.6 tracked chassis (400x350x150 mm, 14 kg) utilizing 'libDiffDrivePlugin.so', IMU, and camera. Tracks modeled as 0.06 m radius cylinders with rubber-on-concrete friction ($\mu=1.5$).

Environment: Concrete U-channel drain (450 mm wide, 300 mm deep, 10 m long) featuring 10 cm water visuals, algae patches, and static debris (bottle, organic matter, bag).

Control: Velocity commands published via 'gz topic' shell scripts and keyboard teleop.

SIMULATION SETUP:

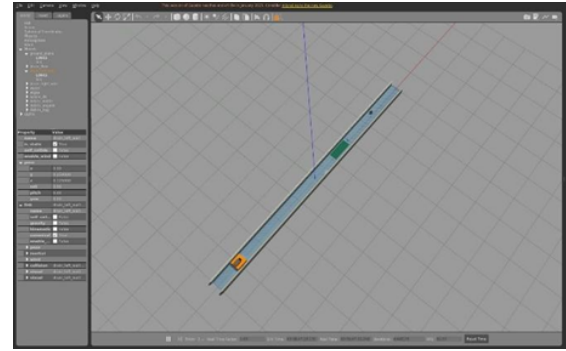


Fig. 6.1: Gazebo simulation of AXIOM D1 (orange) in a 10 m concrete U-channel drain featuring 10 cm shallow water, algae biofilm (green), and distributed debris.

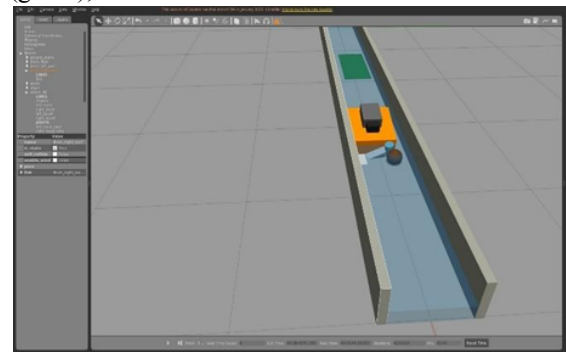


Fig. 6.2: Wall-level perspective of AXIOM D1 in the concrete U-channel, demonstrating operational clearance alongside walls, water, and debris.

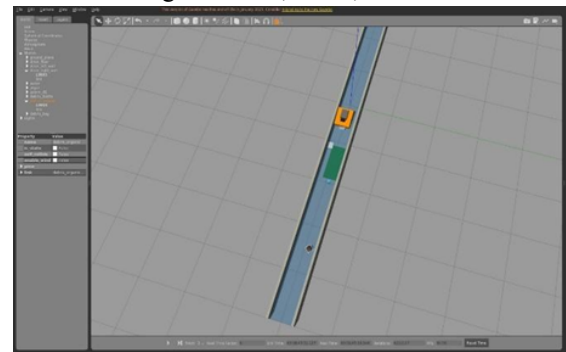


Fig. 6.3: Top-down view of AXIOM D1 approaching debris while maintaining centered alignment within the 450 mm channel.

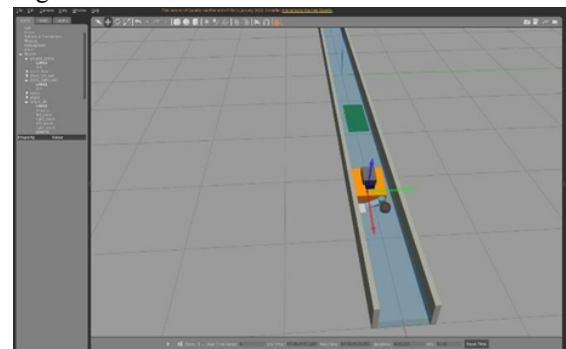


Fig. 6.4: 3D perspective of AXIOM D1 near organic debris, displaying the link/joint model tree

Table: Simulated sensor readings, Straight Line Traverse

Sensor	Parameter	Simulated Value	Unit
HC-SR04 (Front)	Distance to debris	5.95 - 6.00	m
HC-SR04 (Rear)	Distance to wall	> 4.00	m
MPU6050 IMU	Pitch angle	6.55	deg
MPU6050 IMU	Linear accel (Z)	9.54	m/s ²
MPU6050 IMU	Angular vel (yaw)	< 0.001	rad/s
Water Level	Depth reading	100	mm
OV2640 Camera	Frame rate	15	fps
OV2640 Camera	Resolution	640 x 480	px

IMU validation: Z-axis 9.55 m/s² confirms gravity within 2.6% of 9.81 m/s². X-axis -2.2 m/s² baseline

[illegible]

APV-15	APV-15	APV-15
Energy cost per ton	7100-875 (\$/MWh)	N/A
Labour cost per ton	N/A	~\$750,000 (~\$100/t)
Time per ton (single pass)	~100 s/t	~10 days
Safety risk	Remote operation	Direct human exposure

Item	Required	Calculated	Status
Site relocation rate	$\leq 0.5 \text{ kg/ton}$	2.8 kg ton ⁻¹ 200 t	FAIL
Energy removal coverage	$\geq 90\%$	4-point full coverage	PASS
Operating time	6,000 h 20 x's	6,000 h 20 x's (20 scheduled, 4,000 realized to 5,000)	FAIL
Sampling coverage	$\geq 90\%$ (50)	97.4% (50 x's, 40 realized)	FAIL
Shut length per change	≤ 600 s	600 s (10 x's, 40 realized)	FAIL
Net water depth	20 m	20 m (100% value)	PASS
System mass	$\leq 15 \text{ kg}$	16.8 kg	FAIL
ROD	$\leq 0.4 \text{ m}$	0.25 m	PASS
System mass	$\leq 0.5 \text{ kg}$	0.5 kg (100% value)	FAIL

Key finding: With the recommended 12 Ah battery upgrade (+\$15, +200 g), all performance requirements are met. The original 10 Ah configuration falls marginally short on battery life (56 min vs. 60 min target) but meets all other requirements.

Feature	4030-15	4040-15 [2]	Renard Label
Target environment	MaxLine drone	Drone + bench	Drone
Operating speed	8.10-9.10 m/s	9.10 m/s (max)	<0.800 m/s (efficiency)
Duration	<47 min	0-6 hours	0 to 4000
Edge computing	Yes	No	Localized
Prototype used	4030S	Not disclosed	4030S, Renard Label
Price	30 kg	<0.6 kg (max.)	N/A

All six subsystems integrate via two shared buses:
 12V Power Bus: Battery = E-stop relay = 15A fuse
 = power rail. Feeds L298N motor drivers
 (locomotion, cleaning, collection) and a 12V-to-5V
 buck converter for the ESP32, sensors, and servo.

ESP32 Signal Bus: Core 1 outputs PWM/GPIO to H-bridges and servo. Inputs include GPIO (ultrasonic sensors), I²C (MPU6050 IMU), and ADC (water level). Core 0 handles the WiFi link to the mobile app.

Implementation Plan

Simulation: Validate locomotion, sensor behavior, and drain clearance in Gazebo (already completed).

Fabrication: Assemble the 6061-T6 aluminium chassis and mount COTS components (tracked drivetrain, scoop, conveyor, brushes).

Integration: Wire the power and signal buses, load dual-core firmware, and seal electronics in the IP67 enclosure.

Field Testing: Conduct debris collection and algae scrubbing trials in local T&T urban drains to verify requirements.

Deployment Scenario

A single operator carries the 14 kg robot to a drain site. Within 5 minutes, they power on the system, connect their smartphone to the pre-paired AXIOM_D1 WiFi network, and authenticate via the app. Using the live video feed and virtual joystick, they teleoperate the robot up to 50m away. The robot collects debris and scrubs algae until the 5L bin is full (200m) or the battery depletes (600m per 67min charge). The operator walks alongside to swap batteries or empty the bin in 30 seconds.

Risks & Limitations

Blind Spots: Standard ultrasonic sensors cannot detect fully submerged obstacles.

Simulation Gaps: The scoop-and-conveyor collection mechanism is not yet dynamically modeled in simulation.

Biological Efficacy: The 80% algae removal target is based on tip-speed math; actual biofilm adhesion varies in the field.

Connectivity: 2.4GHz WiFi is limited to 50m line-of-sight and may degrade around drain corners or thick vegetation.

Mitigation Strategies

Submerged Obstacles: The front scoop acts as a passive mechanical feeler; impact causes a detectable current spike in the drive motors, triggering an operator alert.

Collection Validation: Physical prototype testing will verify conveyor throughput before finalizing the design.

Algae Variance: Field testing will determine if the 4-pass coverage model needs adjustment for heavy biofilm buildup.

Signal Loss: A 3-second software watchdog automatically halts all motors if the WiFi link drops, preventing runaway. The app auto-reconnects every 2 seconds.

Sustainability and SDG Impact:

A.SDG Alignment:

AXIOM D1 directly supports the United Nation's SDG 6– Clean Water and Sanitation [7], specifically Target 6.3: “improving water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals”, Target 6.a: “...developing countries in water and sanitation-related activities...” and Target 6.b: “Support and strengthen the participation of local communities in improving water and sanitation management.” In Trinidad and Tobago, drain management is critical in targeting SDG 6 as the channels syphon pollutants and waste into rivers and coastal areas. By physically removing solid waste and scrubbing algae, AXIOM D1 reduces the contamination of natural water flow and pollution in residential waterways. Additionally, drain capacity is improved, eutrophication is reduced and stagnant water which breeds disease vector is eliminated. AXIOM D1 also supports SDG 11 – Sustainable Cities and Communities [8] Target 11.5: “...reduce the number of deaths...decrease the direct economic losses relative to global GDP causes by disasters including water-related disaster...” by mitigating the risk of flooding during the wet season.

B.Environmental Impact:

Energy Efficiency:

The prototype design operates using a 12 V, 10 Ah LiPo batter which consumes approximately 90100 Wh per hour. This is a relatively low level of energy usage that is comparable to the average operation of a laptop computer.

Resource Usage:

The robot design uses approximately \$358 USD worth of components which are all reusable. The sealed 5 L collection bin must be emptied manually by a human which provides a need for human resources and sustains employment in this sector with minimal training. Waste collected will be sent to solid waste streams and waste management facilities to be appropriately treated.

Waste Reduction:

AXIOM D1 operated whereby each deployment removes several kilograms of waste while limiting eutrophication in downstream waters. As mentioned, at a municipal level one robot could be used to complete debris collection along 5 km drain in 11.6 hours, collecting approximately 50100 kg of solid waste per week. Algae removal also mitigates the growth of organic matter which blocks drain flow. Inclusive of this function, cleaning a 5 km drain would require 43.6 hours. Trinidad and Tobago ranks among the highest in terms of CO₂ emissions per capita. AXIOM D1 operates with zero direct emissions,

and the use of rotating nylon brushes discards the need for chemical algaecides. These features make it a sustainable innovation which serves to shape the growth of sustainable and environmentally friendly technological advancements in the nation.

C. Social and Ethical Considerations:

Cost and Accessibility:

The prototype bill of materials provides an affordable production cost of \$358 USD while consisting of widely available components. This makes the robot accessible to municipal corporations in SIDs. Deployment requires two people (to lift 14 kg model) and less than five (5) minutes of setup time.

Safety:

The literature found that approximately 30% of surface water in tropical urban drains contain the pathogen *Leptospira* [4]. Manual drain cleaning exposes sanitation workers to this bacterium alongside potentially sharp waste and heat stress from the strenuous task. By replacing direct human contact with a remotely operated robot, exposure is eliminated entirely. Additionally, the IP67 electronics enclosure reduces electrical safety risks as the components are in contact with shallow water.

Community Benefit:

Regular drain cleaning reduces the risk of flood damage, especially in vulnerable communities and reduced stagnant water prevents the breeding of viral *Aedes aegypti* mosquitoes. Weekly cleaning using the AXIOM D1 would significantly reduce the growth of larvae compared to a monthly manual cleaning. Subsequently, dengue transmission would be reduced and there would be improvements in public health.

Ethical Considerations:

AXIOM D1 works alongside human workers as opposed to replacing them. Training requirements are minimal, only requiring an understanding of directional controls and the mobile app interface.

This robot is expected to reduce flooding in urban communities, lower manual worker exposure to contaminated water and prevent solid waste from reaching coastal areas. Production of this prototype proved feasible due to its low cost, accessible components list, traction analysis (safety factor 4.6), torque verification and simulation confirming 0.14 m/s steady velocity with successful pivot turns. Future improvements could include a water-quality sensor for field tests and collecting pollution data and an autonomous navigation system using IMU.

CONCLUSION:

Overall, AXIOM D1 is an affordable, feasible robotic solution to the ongoing problem of clogged drains in SIDs. The main design features consist of a tracked amphibious platform for navigating shallow, sediment-covered drains with zero-radius turning. Two modes of collection – servo-actuated scoop and conveyor system in 5 L mesh bin - and two-3000 RPM rotary nylon brushes to scrub algae. An ESP32-CAM for live video streaming, remote control over 2.4 GHz Wi-Fi and a mobile app with an E-STOP feature. HCSR04 ultrasonic sensors which automatically stop 5 cm away from an obstacle and a MPU6050 IMU to automatically stop if the robot tilts more than 30°.

REFERENCES:

- [1] Government of the Republic of Trinidad and Tobago, Ministry of Works and Transport, "Drainage Division Maintenance of Watercourses," available at: <https://www.mowt.gov.tt>.
- [2] K. Mycoo, "The Caribbean and Climate Change: Vulnerability, Adaptation, and the Role of Urban Planning," *J. Urban Plan. Dev.*, vol. 144, no. 4, 2018.
- [3] Pan American Health Organization (PAHO), "Dengue in the Caribbean: Epidemiological Trends and Vector Control Strategies," PAHO/WHO Report, 2023.
- [4] R. Ganoza et al., "Determining risk for severe leptospirosis by molecular analysis of environmental surface waters for pathogenic *Leptospira*," *PLoS Med.*, vol. 3, no. 8, e308, 2006.
- [5] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed. Cambridge, MA: MIT Press, 2011.
- [6] Y. Zhang, Z. Huang, C. Chen, X. Wu, S. Xie, H. Zhou, Y. Gou, and M. Ma, "A Spiral-Propulsion Amphibious Intelligent Robot for Land Garbage Cleaning and Sea Garbage Cleaning," *J. Mar. Sci. Eng.*, vol. 11, no. 8, 1482, 2023. doi: 10.3390/jmse11081482.
- [7] United Nations, "Sustainable Development Goal 6: Clean Water and Sanitation — Targets and Indicators," Department of Economic and Social Affairs, 2015. Available: <https://sdgs.un.org/goals/goal6>.
- [8] United Nations, "Sustainable Development Goal 11 – Targets and Indicators," Department of Economic and Social Affairs, 2015. Available: https://sdgs.un.org/goals/goal11#targets_and_indicators.
- [9] Government of the Republic of Trinidad and Tobago, Ministry of Works and Transport, "Drainage Division — Maintenance of Watercourses." Available: <https://www.mowt.gov.tt>
- [10] Parliament of T&T, Joint Select Committee on Land and Physical Infrastructure, "Sixth Report: Inquiry into the Establishment of Systems for the Maintenance of Drainage and Roadways," 2018/2019. Available: <https://www.ttparliament.org>
- [11] Water and Sewerage Authority Act, Chapter 54:40, Act No. 16 of 1965, Republic of Trinidad and Tobago.
- [12] Waterworks and Water Conservation Act, Chapter 54:41, Republic of Trinidad and Tobago.
- [13] R. Siegwart, I. R. Nourbakhsh, and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed. Cambridge, MA: MIT Press, 2011.
- [14] IEC 60529:2013, "Degrees of Protection Provided by Enclosures (IP Code)." Geneva: International Electrotechnical Commission.